

JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

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ABSTRACT

Counterfactual explanations (CE) should identify the few input changes that decisively alter a model prediction, but existing methods often change far more features than the explanation requires, weakening both clarity and actionability. We introduce *COLA*, a model- and generator-agnostic post-hoc framework that directly refines any given CE: it computes an optimal-transport (OT) coupling between factual and counterfactual sets and uses this coupling to drive a Shapley-based attribution (p -SHAP) that selects sparse edits while preserving the target effect. Theoretically, OT minimizes an upper bound on the W_1 divergence between factual and counterfactual outcomes, and under mild conditions the refined counterfactuals are guaranteed not to move farther from the factuals than the originals. Empirically, across four datasets, twelve models, and five CE generators, COLA preserves the same target effects with only 26–45% of the original feature edits. On a small-scale benchmark, COLA is near-optimal.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) provides the transparency needed for reliable and trustworthy artificial intelligence systems (Arrieta et al., 2020; Das & Rad, 2020). Within this area, feature attributions (FA) methods such as Shapley values (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) quantify how strongly each input feature contributes to a machine learning (ML) model’s output, thereby exposing the features that drive complex predictions. In healthcare, for example, Shapley values can identify decisive factors such as age and medical history, helping clinicians interpret model decisions (Ter-Minassian et al., 2023; Nohara et al., 2022). Counterfactual Explanations (CE) (Wachter et al., 2017; Guidotti, 2022) offer a complementary view: they specify how input features must change to obtain a different outcome. The large and diverse CE literature (Guidotti, 2022; Verma et al., 2020) makes clear that no single CE generator can serve all use cases. Some methods return one counterfactual for each factual instance, others treat multiple factual instances jointly and produce one or more counterfactuals per observation, while global methods identify dataset-level directions that move factual instances toward a desired model output (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024). Distributional methods go further by treating factual instances as an empirical distribution and seeking a counterfactual distribution with comparable shape and cost (You et al., 2025). Model assumptions vary just as sharply: some CE algorithms require differentiability, whereas others are tailored to tree-based or ensemble models.

Problem Description and Challenges We target a general problem that sits at the intersection of FA and CE, building on the foundations reviewed in Appendix A. Both tools are indispensable for interpretable and accountable AI, but each is incomplete on its own: FA identifies influential features without necessarily producing an actionable plan, while CE can produce actionable changes that include unnecessary or impractical feature edits. The central question is therefore not whether a counterfactual exists, but which limited set of changes preserves its effect.

This problem exposes three concrete challenges, beyond the related settings summarized in Appendix A. First, a universal CE generator is unrealistic because CE objectives are inherently task-specific. Second, a practical refinement method must not depend on restrictive model assumptions such as differentiability or specialized architecture. Third, ordinary feature importance can actively

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$	$\mathbf{z}'$	$\mathbf{z}''$	$\mathbf{y}^*$
$\mathbf{x}$	$\mathbf{z}'$	$\mathbf{z}''$	$\mathbf{y}^*$
200 5 No	250 8 Yes	200 5 No	Yes
150 3 No	150 3 No	150 7 Yes	Yes
100 2 No	350 9 Yes	100 2 No	Yes
150 6 No	150 6 No	350 6 Yes	Yes

Figure 1: [Example: User engagement on an e-commerce platform] A platform seeks to increase user registrations from interaction data, including money spent ($\mathbf{x}$), clicks ($\mathbf{z}'$), and registration status ($\mathbf{y}^*$). In the factual data ($\mathbf{x}$), no users are registered. Action plans $\mathbf{z}'$ and $\mathbf{z}''$ both move user characteristics toward the desired full-registration outcome ($\mathbf{y}^*$). They achieve the same half counterfactual effect, but $\mathbf{z}''$ is the stronger explanation because it requires fewer modifications than $\mathbf{z}'$.

mislead action selection when attribution scores are disconnected from the counterfactual pathway. In particular, it is not sufficient to perform FA independently of CE to select the most important features to change. Result II in Section 6 demonstrates that this decoupling can produce counterproductive user or stakeholder actions because globally important features need not be the features that efficiently achieve the desired counterfactual outcome.

Main Contributions We introduce COUNTERFACTUALS WITH LIMITED ACTIONS (COLA), a general post-hoc framework that decisively refines CE across models and CE generators by using an optimal transport (OT)-induced coupling between factual and counterfactual sets to guide Shapley attributions. This construction yields p -SHAP, a joint-probability-informed Shapley method that unifies commonly used Shapley variants under appropriate couplings and provides a modular interface for attribution-driven edit selection. We establish theoretical guarantees: OT minimizes an upper bound on the W_1 divergence, and under mild conditions the refined counterfactuals are provably no farther from factuals than the originals. Empirically, across four datasets, twelve models, and five CE methods, COLA with p -SHAP reaches the same target effects with only 26–45% of the original feature edits. On a small-scale benchmark, COLA is near-optimal.

To our knowledge, this is the first framework that systematically addresses the problem in Figure 1 without committing to specific CE algorithms or model classes.

2 PROBLEM FORMULATION

We now formalize the sparse counterfactual refinement problem illustrated in Figure 1. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ denote a black-box ML model, and let $\mathbf{x}$ denote observed factual data with n rows and d columns ($\mathbf{x} \in \mathbb{R}^{n \times d}$, $n \geq 1$, and $d \geq 1$). Let $\mathbf{y}^*$ be the target model output ($\mathbf{y}^* \in \mathbb{R}^m$, $m \geq 1$). The task is to find counterfactual data $\mathbf{z}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$) that drives the model output as close as possible to $\mathbf{y}^*$ while obeying a strict budget C on feature changes. Let D denote a divergence function measuring dissimilarity between two entities. The resulting optimization problem is:

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The objective equation (1a) and constraint equation (1b) retain the core CE requirement: $\mathbf{z}$ should make $f(\mathbf{z})$ close to $\mathbf{y}^*$ while remaining close to $\mathbf{x}$. Thus $\mathbf{z}$ serves as the contrastive reference explaining why $f(\mathbf{x})$ does not attain $\mathbf{y}^*$. We deliberately leave D general; it can instantiate Euclidean distance, OT, maximum mean discrepancy (MMD), or differences in mean or median model outcomes. Constraints equation (1c)–equation (1e) add the decisive sparsity requirement. On top of the counterfactual data $\mathbf{z}$, the indicator variable $\mathbf{c}$ determines which entries may change: z_{ik} remains

fixed when $c_{ik} = 0$, and at most C entries can be changed by equation (1c). From equation (1d) and equation (1e), $c_{ik} = 0$ enforces $z_{ik} = x_{ik}$, whereas $c_{ik} = 1$ releases z_{ik} through the sufficiently large constant M .

To solve equation (1), we use FA to identify the feature entries that should determine the modification indicator variable $\mathbf{c}$. The next section introduces Shapley value methods for FA; these baselines, together with our proposed attribution, become the attribution engine inside COLA for constructing the refined counterfactual $\mathbf{z}$. The optimization is computationally difficult even for $d = 1$ with linear models, as shown in Appendix B.

3 PRELIMINARIES ON SHAPLEY VALUE

This section establishes the Shapley-value foundation used throughout COLA. We first recall the game-theoretic Shapley value and then position the feature-attribution variants that serve as baselines for our proposed method. A broader survey of Shapley value methods is available in (Sundararajan & Najmi, 2020).

In cooperative game theory, a coalitional game is defined by a finite set of players—features in our setting— $\mathcal{F} = \{1, 2, \dots, d\}$, and a characteristic function v . The function $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$ assigns each subset $S \subseteq \mathcal{F}$ a payoff $v(S)$, with $v(\emptyset) = 0$. The Shapley value then provides the canonical way to allocate the grand-coalition payoff $v(\mathcal{F})$ across individual players.

Before defining the Shapley value, let $\Delta(k, S)$ denote the marginal contribution of player (feature) k to coalition S . It is the additional payoff obtained when k joins S :

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value ϕ_i for a player i is defined in equation (3).

$$\phi_k = \frac{1}{d} \sum_{S \subseteq \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

The value ϕ_k averages the normalized marginal contributions $\Delta(k, S)$ of feature k over all coalitions excluding k . The inverse binomial factor $\binom{d-1}{|S|}^{-1}$ gives equal weight to coalition sizes, and the final average yields the standard fair and axiomatic payoff allocation across features.

For any data point $\mathbf{x}_i$, $\mathbf{x}_{i,S}$ denotes the entries restricted to features in S , and $\mathbf{x}_{i,\mathcal{F} \setminus S}$ denotes the complementary entries. We use this convention throughout the Shapley variants below.

Baseline Shapley (B-SHAP). This approach uses a specific baseline data point $\mathbf{r}_j$ as the reference for $\mathbf{x}_i$. Its set function is

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j,\mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

where feature absence is modeled by substituting the corresponding value from the reference baseline $\mathbf{r}$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). This method therefore requires an exact alignment between $\mathbf{x}$ and $\mathbf{r}$.

Random Baseline Shapley (RB-SHAP). This B-SHAP variant replaces the fixed baseline with a background distribution and is implicitly used in (Lundberg & Lee, 2017) and the widely used “SHAP” library. Its set function is

$$v_{RB}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

where $\mathcal{D}$ is the background distribution, commonly chosen as the training dataset (Lundberg & Lee, 2017; Merrick & Taly, 2020).

Counterfactual Shapley (CF-SHAP). This method extends RB-SHAP by taking $\mathcal{D}$ to be the counterfactual distribution associated with each factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$).

$$v_{CF}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

This method has been adopted in (Albini et al., 2022; Kommiya Mothilal et al., 2021) and has demonstrated advantages for contrastive FA. Its central requirement is a probabilistic alignment between $\mathbf{x}$ and $\mathbf{r}$.

4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY (p -SHAP) AND ITS THEORETICAL ASPECTS

(Proposed) p -SHAP We generalize equation (4) through equation (6) by introducing an algorithm A_{Prob} that supplies the joint probability linking factual and counterfactual instances.¹ Our p -SHAP is defined as follows:

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_i, S; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

Crucially, unlike prior methods that rely on random baselines, p -SHAP directly addresses the alignment problem by identifying an optimal coupling via OT. This turns feature attribution from a standalone prediction decomposition into a transport-informed explanation problem that explicitly minimizes explanation cost.

Importantly, A_{Prob} is independent of the CE algorithm and only depends on the generated counterfactual $\mathbf{r}$. By conditioning only on these fixed objects, p -SHAP produces consistent FA without inheriting the variability of different CE generation mechanisms; this is the key distinction from CF-SHAP. Contrary to the common intuition that the CE generator’s own pairing should be preferred, Result II in Section 6 shows that OT can be more effective than relying on a counterfactual distribution defined by a CE generation mechanism as done by Albini et al. (2022).

We therefore center the paper on the OT problem, equivalently the entropic 2-Wasserstein formulation below. The resulting transportation plan $\mathbf{p}_{\text{OT}}$ obtained by solving OT becomes the joint distribution of $\mathbf{x}$ and $\mathbf{r}$ used by p -SHAP.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

Here $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ represent the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$, respectively, and $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of feasible joint distributions, i.e., all transport plans. The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ is the entropic regularizer with coefficient $\varepsilon \geq 0$; choosing $\varepsilon > 0$ accelerates OT computation.

Theoretical Aspects of p -SHAP OT minimizes the total feature modification cost (i.e., modifying $\mathbf{x}$ toward $\mathbf{r}$) under the learned alignment between factual $\mathbf{x}$ and counterfactual $\mathbf{r}$. This directly matches the central goal of sparse counterfactual refinement: achieve the counterfactual outcome with minimal modification cost. Theorem 4.1 (proof in Appendix C) makes this connection precise under Lipschitz continuity of f . It establishes that the transportation plan $\mathbf{p}_{\text{OT}}$ used in p -SHAP minimizes an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$. Specifically, the 1-Wasserstein distance between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant times the square root of the minimized expected cost of changing $\mathbf{x}$ toward $\mathbf{r}$, i.e. $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$, where p_{ij} ($j = 1, 2, \dots, m$) specifies how the feature values of $\mathbf{x}_i$ align with those of one or more $\mathbf{r}_j$. Operationally, this means that p -SHAP uses $\mathbf{p}_{\text{OT}}$ as the cost-minimizing strategy for adjusting the feature values of $\mathbf{x}$ toward those of $\mathbf{r}$. Among all feasible modification plans ($\mathbf{p} \in \Pi$), $\mathbf{p}_{\text{OT}}$ achieves the smallest expected modification cost between $\mathbf{x}$ and $\mathbf{r}$ and therefore gives the tightest bound of this form on the counterfactual-effect violation $W_1(f(\mathbf{x}), \mathbf{y}^*)$ in proportion to this cost.

Theorem 4.1 (p -SHAP Towards Counterfactual Effect). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\boldsymbol{\pi} \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{\text{OT}}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

¹First, p -SHAP degrades to B-SHAP in equation (4) when A_{Prob} defines a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that indicates an $i \leftrightarrow j$ alignment of for any $\mathbf{x}_i, \mathbf{r}_j$. Second, p -SHAP degrades to RB-SHAP in equation (5) when A_{Prob} is defined to be independent of CE but associates with an arbitrary distribution $\mathcal{D}$. Third, p -SHAP degrades to CF-SHAP in equation (6) when A_{Prob} is built upon a known distribution of CE.

Thus, $\mathbf{p}_{OT}$ minimizes the expected-feature-cost upper bound on $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Although Theorem 4.1 bounds transport cost rather than the discrete L_0 objective directly, it supplies a principled convex proxy for the $\mathcal{NP}$ -hard sparsity problem. By concentrating attribution mass on efficient transport paths, the OT coupling guides the greedy selection in Algorithm 1 toward discarding non-essential features. Theorem 4.2 further establishes the interventional interpretation of p -SHAP: the attribution corresponds to the effect of feature interventions on the counterfactual reference distribution (proof in Appendix D).

Theorem 4.2 (Interventional Effect of p -SHAP). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Finally, p -SHAP preserves the standard Shapley axioms inherited by B-SHAP and RB-SHAP, reinforcing its validity as a feature-attribution tool. We omit the proof and refer to (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) for the Shapley axioms.

5 THE ALGORITHMIC FRAMEWORK COLA

Sketch The COLA framework in Algorithm 1 operationalizes equation (1) through four modular algorithmic components. We detail Algorithm 1 below and illustrate its execution in Figure 2.

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \mathbb{R}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$
1: Use $A_{CE}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
2: Use $A_{Prob}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
3: Use $A_{Shap}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\phi \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
4: Normalize the element-wise absolute values of ϕ , i.e., $\varphi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$ ($\varphi \in \mathbb{R}_+^{n \times d}$).
5: Use $A_{Value}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$.
6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
7: Sample C pairs (i, k) according to the probability matrix φ , and let $c_{ik} = 1$ for them.
8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
9: **for** $i \leftarrow 1$ to n **do**
10: **for** $k \leftarrow 1$ to d **do**
11: **if** $c_{ik} = 1$ **then**
12: $z_{ik} \leftarrow q_{ik}$
13: **end if**
14: **end for**
15: **end for**
16: **return** $\mathbf{c}$ and $\mathbf{z}$

Line 1 (Applying a CE algorithm to find a counterfactual $\mathbf{r}$). The CE generator A_{CE} takes the model f , the factual $\mathbf{x}$, the target outcome $\mathbf{y}^*$, and the tolerance ϵ as input. It returns a counterfactual $\mathbf{r}$ staying close with $\mathbf{x}$ while attaining $\mathbf{y}^* = f(\mathbf{r})$ and satisfying $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.

Line 2 (Seeking a joint distribution of $\mathbf{x}$ and $\mathbf{r}$). The algorithm A_{Prob} takes $\mathbf{x}$ and $\mathbf{r}$ and returns a matrix encoding the joint distribution over the n and m data points in $\mathbf{x}$ and $\mathbf{r}$, respectively. This matrix $\mathbf{p}$ is the factual-counterfactual alignment used in Line 5 to compute candidate replacement values for $\mathbf{z}$. As discussed in Section 3, when the row alignment is unknown, A_{Prob} can use OT to obtain the joint distribution that minimizes OT distance between $\mathbf{x}$ and $\mathbf{r}$. When trusted alignment information is available, A_{Prob} should encode the alignment between the rows in $\mathbf{x}$ and $\mathbf{r}$ directly.

Lines 3–4 (p -SHAP FA). We apply equation (7) to compute Shapley values for $\mathbf{x}$. The joint distribution $\mathbf{p}$ supplies the alignment between each row of $\mathbf{x}$ and its most relevant counterfactual rows in $\mathbf{r}$, so the

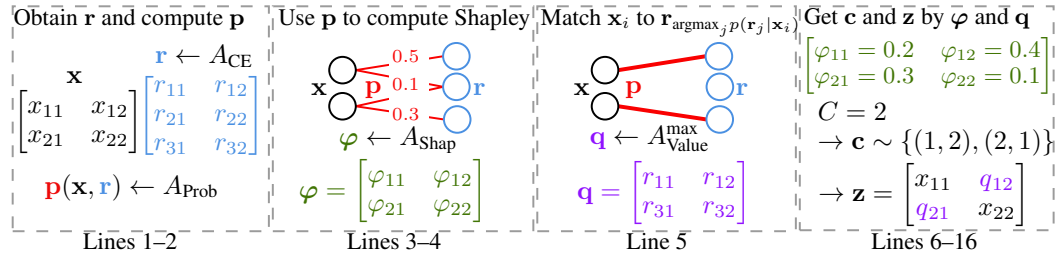


Figure 2: [An illustration of COLA] COLA constructs the action plan $\mathbf{c}$ and refined counterfactual $\mathbf{z}$ for equation (1). For clarity, line 5 is illustrated with $A_{\text{Value}}^{\max}$. In lines 6–16, $C = 2$, so the sampling step selects exactly two modification positions according to the probability matrix $\boldsymbol{\varphi}$.

selected rows in $\mathbf{r}$ become the contrastive references for the corresponding row in $\mathbf{x}$. Numerically, this alignment is a central driver of the resulting FA. The element-wise absolute Shapley values of $\mathbf{x}$ are then normalized to sum to one. The resulting matrix $\boldsymbol{\varphi}$ is the action-selection policy used by COLA.

Line 5 (Computing feature values). The algorithm A_{Value} takes the counterfactual set $\mathbf{r}$ and joint distribution $\mathbf{p}$ as input. For each row i in $\mathbf{x}$, it selects one or more rows in $\mathbf{r}$ as the reference values for modifying $\mathbf{x}$. It returns $\mathbf{q} \in \mathbb{R}^{n \times d}$, where each q_{ik} is the candidate counterfactual replacement for x_{ik} ($\forall i, k$). We use $A_{\text{Value}}^{\max}$ for single-row selection and $A_{\text{Value}}^{\text{avg}}$ for multi-row averaging.

For any row $\mathbf{x}_i$, $A_{\text{Value}}^{\max}$ selects the row of $\mathbf{r}$ with the highest coupling probability.

$$\mathbf{q} = A_{\text{Value}}^{\max}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i),k} \text{ and } \tau(i) = \arg \max_{j=1,2,\dots,m} p_{ij}. \quad (9)$$

The algorithm $A_{\text{Value}}^{\text{avg}}$ computes q_{ik} as a coupling-weighted convex combination of $r_{1k}, r_{2k}, \dots, r_{mk}$.

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16. Because the non-negative matrix $\boldsymbol{\varphi}$ is normalized to sum to one, COLA uses it as a policy for selecting positions in $\mathbf{x}$ for value replacement (i.e., entries with $c_{ik} = 1$), as in line 7. For each selected entry, x_{ik} is replaced by q_{ik} , producing the refined matrix $\mathbf{z}$. The returned pair $(\mathbf{c}, \mathbf{z})$ is the sparse action plan and refined counterfactual for equation (1).

By Theorem 4.1, COLA targets the dissimilarity between $f(\mathbf{z})$ and $\mathbf{y}^*$ through attribution-guided edits that prioritize features most relevant to the desired counterfactual effect. The theorem below (proof in Appendix E) gives the corresponding proximity guarantee: the refined $\mathbf{z}$ produced by COLA satisfies equation (1b) whenever the aligned proposal does, because it does not move farther from the factual input than that aligned counterfactual proposal. Table 3 provides empirical support across the evaluated settings.

Theorem 5.1 (Counterfactual Proximity). Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is no farther from the factual input $\mathbf{x}$ than the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Corollary 5.2. In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

which recovers the concrete guarantee that the refined counterfactual is at least as close to the facts as the original counterfactual set reordered by σ .

Complexity of COLA Let $\mathcal{O}(M_{\text{CE}})$ denote the complexity of A_{CE} . For A_{Shap} , consider weighted linear regression for Shapley estimation and let M_{Shap} be the number of sampled subsets. The complexity of COLA with respect to n, m, d , and the entropic OT regularization parameter ε is $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ for $A_{\text{Value}}^{\text{max}}$ and $N = \mathcal{O}(nmd)$ for $A_{\text{Value}}^{\text{avg}}$. Appendix F gives the full derivation.

6 NUMERICAL RESULTS

This section evaluates how decisively COLA solves equation (1), with $\mathbf{y}^* = f(\mathbf{r})$ where $\mathbf{r}$ is the counterfactual returned by a CE method A_{CE} . We use four divergence functions: OT measures distributional distance, MMD measures divergence between distribution means in a high-dimensional feature space, and absolute mean difference (MeanD) and absolute median difference (MedianD) measure mean and median differences, respectively. The numerical evidence establishes three points: I) *COLA’s effectiveness for modification minimality*. II) *p-SHAP’s superior performance over other Shapley methods for modification minimality*. III) *COLA’s near-optimal performance*.

Experiment Setup The experiments cover 4 binary-classification datasets, 5 CE algorithms designed for distinct goals, and 12 classifiers, as listed in Table 1. Each combination of dataset, A_{CE} algorithm, and model defines one “experiment scenario”.

Table 1: [Experiment Scenarios Setup] The benchmark spans four datasets, five CE algorithms, and 12 models. A “scenario” is one combination of dataset, A_{CE} algorithm, and model f .

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: [Experiment Methods Setup] Six comparison methods isolate the roles of coupling, value assignment, and Shapley attribution; colors match Figure 3. Each method is evaluated within the scenarios defined in Table 1.

Method	The probability $\mathbf{p}$ used by A_{Value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB-p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB-p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8) (but not used in } A_{\text{Shap}})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF-p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF-p_{Rnd}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF-p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF-p_{Ect}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

We use the diverse A_{CE} methods in Table 1 to test COLA beyond any single CE design. DiCE and KNN are instance-based CE methods that generate counterfactuals for individual factual instances. AReS and GlobeCE are group-based methods that produce counterfactual collections for the factual data as a group. Discount treats the factual instances as an empirical distribution and seeks a nearby counterfactual distribution.

Table 2 defines six methods, with CF- p_{OT} representing the proposed p -SHAP configuration and the remaining methods serving as controlled baselines. We benchmark each method across the scenarios in Table 1. Each method is specified by the triplet A_{Prob} , A_{Value} , and A_{Shap} . For example, RB- p_{Uni} uses a uniform A_{Prob} to compute $\mathbf{p}$; this $\mathbf{p}$ is then used in equation (9) or equation (10) to compute $\mathbf{q}$, while A_{Shap} is RB-SHAP.

These methods form targeted ablations. First, $\text{RB-}p_{\text{Uni}}$ and $\text{CF-}p_{\text{Uni}}$ isolate the value of CE information because only the latter uses it. Second, $\text{CF-}p_{\text{Uni}}$, $\text{CF-}p_{\text{Rnd}}$, and $\text{CF-}p_{\text{OT}}$ all use CE information but differ in the joint distribution, allowing us to show that the OT-derived distribution is the strongest alignment. Third, comparing $\text{CF-}p_{\text{OT}}$ with $\text{RB-}p_{\text{OT}}$ confirms that OT matters because it informs A_{Shap} through factual-counterfactual alignment, not merely because it appears elsewhere in the pipeline. Finally, $\text{CF-}p_{\text{Ect}}$ covers the special case where each counterfactual has a known source, yielding an exact factual-counterfactual alignment and making CF-SHAP coincide with B-SHAP.

Table 3: [COLA for Modification Minimality] Number of modified features in $\mathbf{z}$ required by each method when $f(\mathbf{z})$ reaches 80% of the counterfactual effect of $f(\mathbf{r})$. Each entry averages 100 runs in each of 4 randomly selected scenarios from Table 1. The symbol “-” indicates that the target counterfactual effect is not achieved.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	$\text{RB-}p_{\text{Uni}}$	—	—	—	—
	$\text{RB-}p_{\text{OT}}$	5.29(± 0.09)	75.9%	—	—
	$\text{CF-}p_{\text{Uni}}$	—	—	—	—
	$\text{CF-}p_{\text{Rnd}}$	—	—	—	—
	$\text{CF-}p_{\text{OT}}$	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	$\text{RB-}p_{\text{Uni}}$	7.10(± 0.08)	24.5%	—	—
	$\text{RB-}p_{\text{OT}}$	8.55(± 0.08)	50.2%	—	—
	$\text{CF-}p_{\text{Uni}}$	7.01(± 0.07)	41.1%	—	—
	$\text{CF-}p_{\text{Rnd}}$	10.63(± 0.08)	62.4%	—	—
	$\text{CF-}p_{\text{OT}}$	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	$\text{RB-}p_{\text{Uni}}$	5.02(± 0.05)	82.7%	—	—
	$\text{RB-}p_{\text{OT}}$	—	—	—	—
	$\text{CF-}p_{\text{Uni}}$	2.80(± 0.04)	34.4%	—	—
	$\text{CF-}p_{\text{Rnd}}$	2.58(± 0.04)	32.1%	—	—
	$\text{CF-}p_{\text{OT}}$	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	$\text{RB-}p_{\text{Uni}}$	—	—	—	—
	$\text{RB-}p_{\text{OT}}$	—	—	—	—
	$\text{CF-}p_{\text{Uni}}$	—	—	—	—
	$\text{CF-}p_{\text{Rnd}}$	2.73(± 0.04)	15.7%	—	—
	$\text{CF-}p_{\text{OT}}$	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

Result I: COLA achieves significant action reduction with a minor loss in counterfactual effect

In Table 3, we require $\mathbf{z}$ to reach either 80% or 100% of the counterfactual effect of $f(\mathbf{r})^2$. The results show that COLA reaches these goals with far fewer actions in $\mathbf{z}$ (i.e., modifications of features in $\mathbf{z}$) than the original CE $\mathbf{r}$. For the 80% target, COLA requires only 13%–25% of the original feature changes, measured by $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$. For the 100% target, only COLA with p -SHAP ($\text{CF-}p_{\text{OT}}$) succeeds, using only 26%–45% of the feature changes in the original $\mathbf{r}$. Thus p -SHAP delivers the strongest action minimality for the sparsity in $\mathbf{z}$ among the evaluated methods.

Result II: p -SHAP outperforms the other Shapley methods in achieving counterfactual effect

Figure 3 compares the Shapley methods in equation (4)–equation (7), with the number of allowed feature changes C on the x-axis and $D(f(\mathbf{z}), \mathbf{y}^*)$ from equation (1) on the y-axis. First, $\text{RB-}p_{\text{Uni}}$ and $\text{RB-}p_{\text{OT}}$ perform markedly worse than the CE-informed methods, establishing that sparse CE refinement requires counterfactual information in the attribution step. Second, $\text{CF-}p_{\text{OT}}$ outperforms $\text{RB-}p_{\text{OT}}$, demonstrating that OT improves p -SHAP specifically by providing effective factual-counterfactual alignment, rather than through unrelated pipeline effects, because the two methods differ only in A_{Shap} (see Table 2). Third, p -SHAP substantially outperforms $\text{CF-}p_{\text{Uni}}$ and $\text{CF-}p_{\text{Rnd}}$, confirming that counterfactual information alone is insufficient: the factual-counterfactual alignment must be

²Note that by definition, we have $D(f(\mathbf{r}), \mathbf{y}^*) = 0$, which represents a 100% counterfactual effect since $\mathbf{y}^* = f(\mathbf{r})$. To define a counterfactual effect 80%, consider the proportion of divergence reduced by the refined CE $\mathbf{z}$. That is, $\text{Counterfactual Effect} = 1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$, with D being OT.

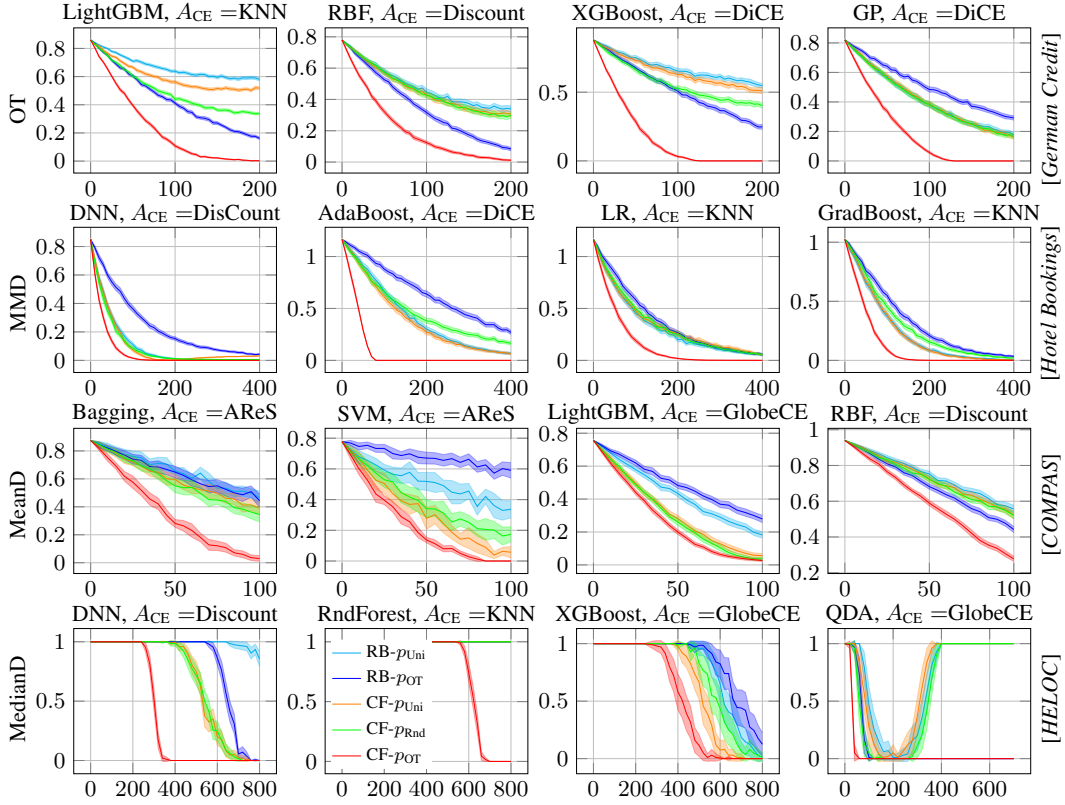


Figure 3: $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the allowed actions C . Each curve uses 100 runs, and shaded regions report 99.9% confidence intervals. $A_{\text{value}}^{\text{avg}}$ is used for *HELOC* and *COMPAS*, and $A_{\text{value}}^{\text{max}}$ is used for *German Credit* and *Hotel Bookings*. The legend inside “RnDForest, $A_{\text{CE}} = \text{KNN}$ ” applies to all plots.

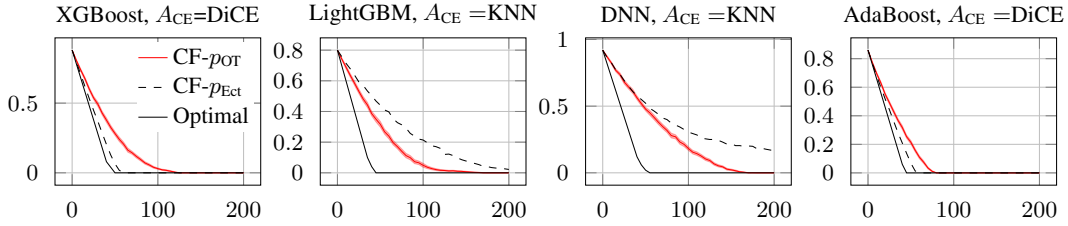


Figure 4: [*German Credit*] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of allowed actions C , using *MeanD* as D .

informative. As Figure 4 further shows, this alignment need not be the exact alignment induced by the original CE generation mechanism.

Across the scenarios in Table 1, COLA with p -SHAP consistently delivers strong performance, as shown in Figure 3 and Table 3. Appendix H reports the broader set of supporting results. Overall, p -SHAP is the strongest Shapley method evaluated for sparse CE refinement.

Result III: COLA is competitive with the mixed integer linear programming (MILP) optimum on tractable benchmarks

This result demonstrates that p -SHAP can reduce dependence on the CE generation process by replacing CE-algorithm-dependent knowledge³ of $\mathcal{D}$ with the OT joint distribution between factual and counterfactual data, as shown in Figure 4. We benchmark CF- p_{Ect} within COLA and solve equation (1) using *MeanD* as the divergence function D . Because CF- p_{Ect}

³This knowledge, i.e. exact alignment between factual and counterfactual, is available only by DiCE and KNN among all CE methods considered.

relies on known factual-counterfactual alignment, it tests how effectively COLA uses exact alignment information. The optimal solution for this tractable setting can be obtained by an MILP; Appendix G derives the formulation. Since solving the MILP is computationally expensive, this benchmark is run only for *German Credit*.

CF- p_{Ect} is near-optimal with DiCE. Because DiCE and KNN produce independent factual-counterfactual pairs, this result confirms that COLA is effective for instance-based CE even though equation (1) also covers group and distributional CE. Notably, CF- p_{OT} sometimes outperforms CF- p_{Ect} , because it uses a theoretically grounded alignment to identify key modifiable features, whereas CF- p_{Ect} relies on CE-algorithm-dependent alignment whose effectiveness for FA is not guaranteed. A gap remains between CF- p_{Ect} , CF- p_{OT} , and the KNN optimum, indicating that identifying the best alignment remains an important open question.

7 CONCLUSIONS

This paper introduced COLA, a general framework for refining CE with joint-distribution-informed Shapley values. By coupling factual and counterfactual sets and using the resulting p -SHAP attribution to select sparse edits, COLA preserves counterfactual effect while requiring substantially fewer actions. Theoretical results justify the OT coupling and proximity of the refined counterfactuals, and experiments across datasets, models, and CE generators show that the framework consistently improves action minimality without committing to a specific CE method or model class.

ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.⁴ This paper develops *COLA* with p -SHAP for refining CE. The contribution is methodological and empirical; it involves no human subjects or interventions, and we did not collect, annotate, or release any new personal data.

All experiments use standard, de-identified, publicly available tabular datasets (e.g., Adult/Census Income, German Credit, HELOC). We follow dataset licenses and terms and use these data solely for benchmarking. We do not attempt re-identification or linkage, and we release no additional personal information. Our code will include options to exclude protected attributes and enforce feasibility and immutability constraints on features.

CE can expose biases in underlying models, but they do not by themselves guarantee fairness. Our method reduces the number of recommended feature edits; it does *not* authorize edits to sensitive or immutable attributes (e.g., sex, race, age), nor does it establish causal validity. We discourage using CE to “game” high-stakes systems (e.g., lending, hiring, healthcare) or to pressure individuals into unrealistic or unethical behavior changes. Any deployment in consequential domains must include domain-expert feasibility constraints, fairness audits (e.g., disparate impact or benefit across groups), human oversight, and compliance with applicable data-protection and AI regulations. Our attribution is distributional rather than causal and must not be interpreted as causal effect.

The main risks are (i) misinterpreting attributions as causal, (ii) recommending infeasible or harmful edits, and (iii) enabling strategic manipulation. We mitigate these risks by (a) stating the non-causal scope and limitations of the method, (b) supporting cost and feasibility masks and immutable features in the implementation, (c) reporting subgroup performance where relevant, and (d) documenting responsible-use guidance.

We provide an anonymized implementation, detailed settings, and seeds to support replication. We report model types, hyperparameters, and compute budgets, and we use fixed evaluation protocols to avoid result cherry-picking. To our knowledge, there are no conflicts of interest or sensitive sponsorships related to this submission; any funding disclosures will be provided after the double-blind review.

⁴<https://iclr.cc/public/CodeOfEthics>

REPRODUCIBILITY STATEMENT

We provide the components needed to reproduce the method and experiments. The COLA framework and p-SHAP attribution are specified by pseudocode and notation in the main paper (Algorithm 1; Method and Preliminaries sections), and the appendix gives the theoretical assumptions and complete proofs.

Appendix I documents hyperparameters, optimization settings, environment specifications, random seeds, datasets, data splits, preprocessing, and evaluation protocols. The experimental sections and ablations enumerate the baselines, CE generators, models, and coupling choices, with configurations mirrored in the supplemental materials. The experiment code and software will be released upon acceptance; Appendix I also reports hardware and runtime, and the appendix includes ablation and sensitivity studies plus the small-scale MILP benchmark setup for independent verification.

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APPENDICES

A	Comparison to Existing Approaches	14
B	$\mathcal{NP}$-hardness of the Problem in equation (1)	14
C	Proof of Theorem 4.1: p-SHAP Towards Counterfactual Effect	15
D	Proof of Theorem 4.2: Interventional Effect	17
E	Proof of Theorem 5.1: Counterfactual Proximity	17
F	Analysis of Computational Complexity of COLA	18
G	An MILP Formulation of equation (1) With MeanD	19
H	Extended Numerical Results	20
I	Experiments Reproducibility	20

A COMPARISON TO EXISTING APPROACHES

(Albini et al., 2022) proposed CF-SHAP, which uses counterfactual data points as the Shapley background distribution. Its strength also exposes its limitation: it assumes that the counterfactual distribution is defined conditionally for each factual instance, which requires a known probabilistic alignment between factual and counterfactual instances. Similar assumptions appear in (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021). This assumption breaks down in many global-explanation settings. The counterfactual-distribution setup of (Albini et al., 2022) is therefore a special case of our framework rather than a substitute for it. WeightedSHAP (Kwon & Zou, 2022) weights features instead of treating them as equally contributing; our method instead weights contributions across data points and can naturally extend to simultaneous row and column weighting. (Kommiya Mothilal et al., 2021) combine FA and CE for explainability, but their CE-based FA retains the same alignment assumption as (Albini et al., 2022), limiting its transfer to group (Ley et al., 2023; 2022) and distributional CE (You et al., 2025). (Sharma et al.) model counterfactuals through causal structure and time-series predictors to decompose a single system-level metric change into input contributions. These lines of work do not solve the minimal-action CE problem addressed here.

The formulations in (Kanamori et al., 2022; Karimi et al., 2021) are closest to our target problem. (Karimi et al., 2021) minimize action cost under a known structural causal model (SCM), but such structure is rarely available in practice. (Kanamori et al., 2022) explicitly note that whether existing CE methods can solve this problem remains open, and propose MILP-solver-based approaches for linear classifiers, tree ensembles, and deep ReLU networks, building on (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). The limitation is decisive: exact MILP solving is expensive and difficult to scale, motivating our post-hoc and model-agnostic refinement approach.

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 establishes that a scalable exact algorithm for the general problem should not be expected unless $\mathcal{P} = \mathcal{NP}$. The difficulty is not merely the possible nonlinearity of f ; it is already present in the combinatorial structure of sparse action selection.

Theorem B.1. *Problem equation (1) is generally $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when $d = 1$ for linear models.*

Proof. Consider the *Sparse Regression (SR) with a Cardinality Constraint* problem. Given a matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and a sparsity level $K \in \mathbb{N}$, the task is to find a vector $\mathbf{z} \in \mathbb{R}^n$ that minimizes residual error while using at most K nonzero elements:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ counts the nonzero elements in $\mathbf{z}$. This problem is known to be $\mathcal{NP}$ -hard because selecting the active subset of variables is combinatorial.

We reduce this SR problem to problem equation (1). Set the number of features to $d = 1$ and the number of instances to n , matching the dimension of the sparse-regression variable. Let the factual data be $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$, let the target model output be the same $\mathbf{y}^* \in \mathbb{R}^m$, and define the model as the linear map $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$.

For the model output, set $D(f(\mathbf{z}), \mathbf{y}^*) = \|\mathbf{f}(\mathbf{z}) - \mathbf{y}^*\|_2^2$, the squared Euclidean distance. For the instances, set $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$, since $\mathbf{x} = \mathbf{0}$. Set the maximum number of allowed feature changes C equal to k , the sparsity level in the SR problem. The large constant M can be any sufficiently large positive number, for example $M \geq \max_i |z_i|$.

Since $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) simplify to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + Mc_i = Mc_i, \\ z_i &\geq 0 \cdot (1 - c_i) - Mc_i = -Mc_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

Thus $c_i = 0$ forces both $z_i \leq 0$ and $z_i \geq 0$, so $z_i = 0$. If $c_i = 1$, then $z_i \in [-M, M]$. Constraint equation (1c) becomes $\sum_{k=1}^n c_i \leq k$.

Under this reduction, problem equation (1) becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \quad (11a)$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \quad (11b)$$

$$\sum_{k=1}^d c_i \leq k, \quad (11c)$$

$$z_i = 0, \quad \text{if } c_i = 0, \quad (11d)$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \quad (11e)$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \quad (11f)$$

Here c_i decides whether z_i may change. The constraints enforce $z_i = 0$ when $c_i = 0$, exactly mirroring the sparse-regression cardinality constraint. The bound $\sum_{k=1}^n c_i \leq k$ permits at most k changes, and the objective is identical to the SR objective.

Because this construction directly embeds cardinality-constrained SR, solving Problem equation (1) is at least as hard as solving an $\mathcal{NP}$ -hard problem. Therefore, Problem equation (1) is $\mathcal{NP}$ -hard even when $d = 1$, the model f is linear, and the divergences D are standard Euclidean distances. $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ be factual data with probability weights $\mu_i \geq 0$ satisfying $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ be counterfactual data with probability weights $\nu_j \geq 0$ satisfying $\sum_{j=1}^m \nu_j = 1$. Let $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ be the OT plan between $\mathbf{x}$ and $\mathbf{r}$ that minimizes expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions satisfying the marginal constraints $\sum_{j=1}^m p_{ij} = \mu_i$ for all i and $\sum_{i=1}^n p_{ij} = \nu_j$ for all j . The theorem below formalizes why this plan aligns feature attributions with expected modification costs, producing action plans that are cost-efficient for achieving counterfactual outcomes.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\boldsymbol{\pi} \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{\text{OT}}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Thus $\mathbf{p}_{\text{OT}}$ minimizes the expected-feature-cost upper bound on $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Proof. Since the model f is Lipschitz continuous with constant L , for any $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{r}_j \in \mathbb{R}^d$, it holds that:

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying both sides of the inequality by $p_{ij}^{\text{OT}} \geq 0$, we obtain:

$$p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing both sides over all $i = 1, \dots, n$ and $j = 1, \dots, m$, we have:

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Let us denote $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$. The inequality then becomes:

$$E_f \leq L E_d.$$

To further bound E_d , we apply the Cauchy-Schwarz inequality. Observe that the weights p_{ij}^{OT} are non-negative and satisfy $\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} = 1$ because $\mathbf{p}_{\text{OT}}$ is a probability distribution over the joint space of $\mathbf{x}$ and $\mathbf{r}$. The Cauchy-Schwarz inequality states that for any real-valued functions a_{ij} and b_{ij} ,

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Setting $a_{ij} = \sqrt{p_{ij}^{\text{OT}}}$ and $b_{ij} = \sqrt{p_{ij}^{\text{OT}}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, we have:

$$E_d = \sum_{i,j} p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{\text{OT}}} \sqrt{p_{ij}^{\text{OT}}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying the Cauchy-Schwarz inequality:

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{\text{OT}} \right) \left(\sum_{i,j} p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since $\sum_{i,j} p_{ij}^{\text{OT}} = 1$, this simplifies to:

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking the square root of both sides, we obtain:

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting back into the inequality for E_f , we have:

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Note that the 1-Wasserstein divergence is no more than E_f ⁵, then,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Because both π and $\mathbf{p}_{\text{OT}}$ denote joint probability of $\mathbf{x}$ and $\mathbf{r}$, however, π makes the summation the minimum for the W_1 term across all possible joint distributions, whereas $\mathbf{p}_{\text{OT}}$ does not.

Therefore, the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant L times the square root of the expected transportation cost under $\mathbf{p}_{\text{OT}}$. Since $\mathbf{p}_{\text{OT}}$ minimizes the expected transportation cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible transport plans in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$, we have

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

Hence the conclusion follows. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the interventional difference between the expected value of $f(\mathbf{r})$ after setting features S and the unconditional expected value of $f(\mathbf{r})$. This relationship is:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ be the joint probability of $\mathbf{x}$ and $\mathbf{r}$ returned by A_{Prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are fixed to $\mathbf{x}_{i,S}$, while the features in $\mathcal{F} \setminus S$ remain distributed according to their marginal distribution $p(\mathbf{r}_{\mathcal{F} \setminus S})$. Therefore, the expected value of $f(\mathbf{r})$ under the intervention is:

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

By the definition of $v^{(i)}(S)$,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

which gives

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

This proves the claim. $\square$

The value function $v^{(i)}(S)$ therefore captures the expected value of $f(\mathbf{r})$ when we intervene by setting the features in S to $\mathbf{x}_{i,S}$, denoted as $do(\mathbf{r}_S = \mathbf{x}_{i,S})$. This intervention does not require a predefined true data-generating joint distribution $p(\mathbf{x}, \mathbf{r})$; it uses the coupling supplied to the attribution procedure. Consequently, $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ gives the base expected value of $f(\mathbf{r})$ plus the interventional contribution assigned to S .

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is no farther from the factual input $\mathbf{x}$ than the aligned counterfactual proposal $\mathbf{q}$, ensuring that refinement introduces no additional divergence.

Proof. Let $\mathbf{q}$ be the aligned reference matrix generated by $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$. For modified entries with $c_{ik} = 1$, $z_{ik} = q_{ik}$. For unchanged entries with $c_{ik} = 0$, $z_{ik} = x_{ik}$. Therefore,

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms with respect to $\mathbf{q}$ and $\mathbf{z}$ are:

$$\begin{aligned}\|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2.\end{aligned}$$

And,

$$\begin{aligned}\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k):c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k):c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0,\end{aligned}$$

where the equality (i) holds because of equation (12).

Because $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, we have:

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots yields:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This proves the proximity guarantee. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

which recovers the guarantee that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Proof. Under $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ is a permutation matrix associated with a bijection σ . Therefore, the value-assignment algorithm A_{Value} , whether maximizing or averaging, assigns $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$ for every instance i . Substituting the permuted matrix $\mathbf{r}_\sigma$ for $\mathbf{q}$ in the general bound from Theorem 5.1, namely $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$, directly proves the result. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

We analyze the computational complexity of COLA component by component.

First, consider A_{Prob} . If the alignment between $\mathbf{x}$ and $\mathbf{r}$ is known a priori, A_{Prob} constructs $\mathbf{p}$ from that knowledge in $\mathcal{O}(n \times m)$. Otherwise, solving OT with the Sinkhorn–Knopp algorithm obtains $\mathbf{p}$ in $\mathcal{O}(n \times m \times \log(1/\varepsilon))$.

Next, consider A_{Shap} . For each sampled subset, the model is evaluated on all n data points, giving $\mathcal{O}(n)$ evaluations per subset. Incorporating reference values from the reference data $\mathbf{r}$ replaces selected feature values by their aligned baselines from the reference table $\mathbf{r}$. This access is $\mathcal{O}(m)$ when reading from the reference table and can be treated as $\mathcal{O}(d)$ when reference values are precomputed and accessed in constant time. With M_{Shap} sampled subsets, the resulting complexity of A_{Shap} is $\mathcal{O}(n \times d \times M_{\text{Shap}})$.

The normalization in line 4 takes $\mathcal{O}(n \times d)$.

For value assignment, $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$ have different costs. For $A_{\text{Value}}^{\text{max}}$, each row $\mathbf{x}_i$ requires (1) computing probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, costing $\mathcal{O}(m)$, (2) identifying the reference

row $\mathbf{r}_j$ with largest p_{ij} , costing $\mathcal{O}(m)$, and (3) assigning $q_{ik} = r_{\tau(i),k}$ where $\tau(i) = \arg \max_j p_{ij}$, costing $\mathcal{O}(d)$. Across n rows, the total complexity is $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

For $A_{\text{Value}}^{\text{avg}}$, each row $\mathbf{x}_i$ requires 1) computing p_{ij} for all $j \in \{1, 2, \dots, m\}$, costing $\mathcal{O}(m)$, 2) computing $\sum_{j'=1}^m p_{ij'}$, also costing $\mathcal{O}(m)$, and 3) calculating the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, costing $\mathcal{O}(m \times d)$ per row. Across n rows, the total complexity is $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, the complexity is $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$ because $C \leq n \times d$.

Therefore, the complexity of COLA with $A_{\text{Value}}^{\text{max}}$ is

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and the complexity of COLA with $A_{\text{Value}}^{\text{avg}}$ is

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Hence the overall complexity of COLA, as a function of n , m , d , and the entropic OT regularization parameter ε , is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ for $A_{\text{Value}}^{\text{max}}$ and $N = \mathcal{O}(nmd)$ for $A_{\text{Value}}^{\text{avg}}$.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

This section provides a global optimality benchmark for the known-alignment case of equation (1) with D instantiated as MeanD:

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

with $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Under COLA, $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$ and $\mathbf{z}$ remains no farther from $\mathbf{x}$ than $\mathbf{r}$, so equation (1b) can be dropped in this benchmark. The resulting formulation of equation (1) is:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

The original constraints equation (1d) and equation (1e) merge into equation (13c) because CF- p_{Ect} is imposed. For each $\mathbf{x}_i$, an exact $\mathbf{r}_j$ serves as its reference in A_{Shap} and A_{Value} , so each x_{ik} ($k = 1, 2, \dots, d$) either stays unchanged or changes to r_{jk} . Thus $z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik})$, with the binary variable c_{ik} determining the chosen value.

Because the alignment between each $\mathbf{x}_i$ and its corresponding $\mathbf{r}_j$ is known, $\mathbf{q}$ is fixed; both $A_{\text{Values}}^{\text{max}}$ and $A_{\text{Values}}^{\text{avg}}$ return the same $\mathbf{q}$. For any data point i and feature set $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denote the candidate $\mathbf{z}_i$ in which all features $k \in S$ are changed to q_{ik} and all features $h \in \mathcal{F} \setminus S$ remain x_{ih} . The collection of $\mathbf{z}_{iS}$ for all $S \subseteq \mathcal{F}$ enumerates the possible values of $\mathbf{z}$. Define the corresponding scalar for each $\mathbf{z}_{iS}$:

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

Then, for any $\mathbf{z}_i$ in equation (13), the term $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$ can be represented using binary variables a_{iS} and scalars g_{iS} :

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

Thus equation (13) becomes the following mixed integer program.

$$\min_{\mathbf{a}, \eta} \quad \eta, \quad (14a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \quad (14b)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \quad (14c)$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \quad (14d)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \quad (14e)$$

Minimizing η under equation (14b) and equation (14c) is equivalent to minimizing the objective of equation (13). Constraint equation (14d) ensures that each data point i selects exactly one feature-modification plan a_{iS} for some $S \subseteq \mathcal{F}$, and equation (14e) enforces the action budget from equation (13b). Solving equation (14) gives the exact benchmark for COLA under known factual-counterfactual alignment, as reported in Figure 4 of Section 6.

H EXTENDED NUMERICAL RESULTS

Figures 5–8 reinforce the main conclusion: CF- p_{Uni} generally outperforms RB- p_{Uni} , and CF- p_{OT} consistently outperforms RB- p_{OT} even though each pair differs only in A_{Shap} . Thus *RB-SHAP is not suitable for FA in CE*.

Lines 3–4 isolate the effect of the Shapley method used for FA. The comparison is decisive in two matched settings. First, RB- p_{Uni} and CF- p_{Uni} differ only in A_{Shap} , and CF- p_{Uni} performs better in Figures 5–8. Second, RB- p_{OT} and CF- p_{OT} also differ only in A_{Shap} , and CF- p_{OT} consistently dominates. These results confirm that CE-aware Shapley attribution is essential for sparse CE refinement.

Beyond FA, line 5 is equally central: the joint probability $p(\mathbf{x}, \mathbf{r})$ constructs $\mathbf{q}$, which determines the direction in which factual features should move toward the target outcome. In Figures 5–8, CF- p_{OT} consistently outperforms all other methods. CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all provide candidate joint probabilities $\mathbf{p}$ when exact factual-counterfactual alignment is unknown, yet their performance differs substantially. This confirms that merely knowing the CE and its marginal distribution is insufficient.

OT is especially valuable when factual-counterfactual alignment is missing or inaccurate. Even when a CE algorithm explicitly matches each factual instance to a counterfactual, that alignment is not necessarily optimal for sparse refinement, as supported by Figure 4 in Section 6.

Importantly, p_{OT} need not be the true data-generating joint distribution of $\mathbf{x}$ and $\mathbf{r}$. Its role is to guide COLA in jointly using $\mathbf{x}$ and $\mathbf{r}$ for both FA and CE. The QDA column in Figure 5 further shows the stability of OT-based methods, while other alignments diverge substantially from the target. At the same time, COLA is *not limited to using OT* as A_{Prob} . As Figure 4 indicates, even strong known choices of $\mathbf{p}$ can retain a non-negligible gap to the global optimum, making better A_{Prob} design a clear direction for future work.

I EXPERIMENTS REPRODUCIBILITY

The experiments were conducted on a high performance computing (HPC) cluster using four nodes in parallel, one for each dataset. Each node has two Intel Xeon Processor 2660v3 CPUs (10 cores, 2.60GHz) and 128 GB memory. Runtime is approximately 5–10 hours per node, depending on dataset size. The experiments can also be reproduced on a laptop, although with longer runtime than on the HPC cluster.

For all four datasets, numerical features are standardized and categorical features use either label encoding or one-hot encoding. In practice, the two encoding choices do not materially change COLA’s performance. The train-test split is 7 : 3.

The optimality baseline in Figure 4 is solved with Gurobi 11.0.2 ([gur](#)). Reproducing this baseline requires a Gurobi license, although open-source operations-research libraries such as Google OR-Tools ([goo](#)) can be used as alternatives. The MILP in Appendix G is computationally expensive and is therefore practical mainly for small-scale datasets such as German Credit; computing the baseline for larger datasets requires reducing the number of features.

The model hyperparameters are fixed as follows. Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA use scikit-learn implementations ([skl](#)) with default hyperparameters. DNN, SVM, RBF, and LR are implemented in PyTorch ([Paszke et al., 2019](#)); the DNN has three layers and the SVM uses a linear kernel. XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) use their scikit-learn interfaces with default hyperparameters.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

We used large language models (LLMs) strictly as general-purpose writing-assistance tools. Their role was limited to grammar, wording, and light copy-editing of author-written text. LLMs did not contribute research ideation, modeling choices, experimental design, or results. All algorithms, proofs, datasets, and analyses were created and verified by the authors. Every LLM-suggested phrase was reviewed and edited before inclusion in the paper. The authors take full responsibility for all content; LLMs are not authors or contributors. This disclosure complies with the ICLR policy on LLM usage.

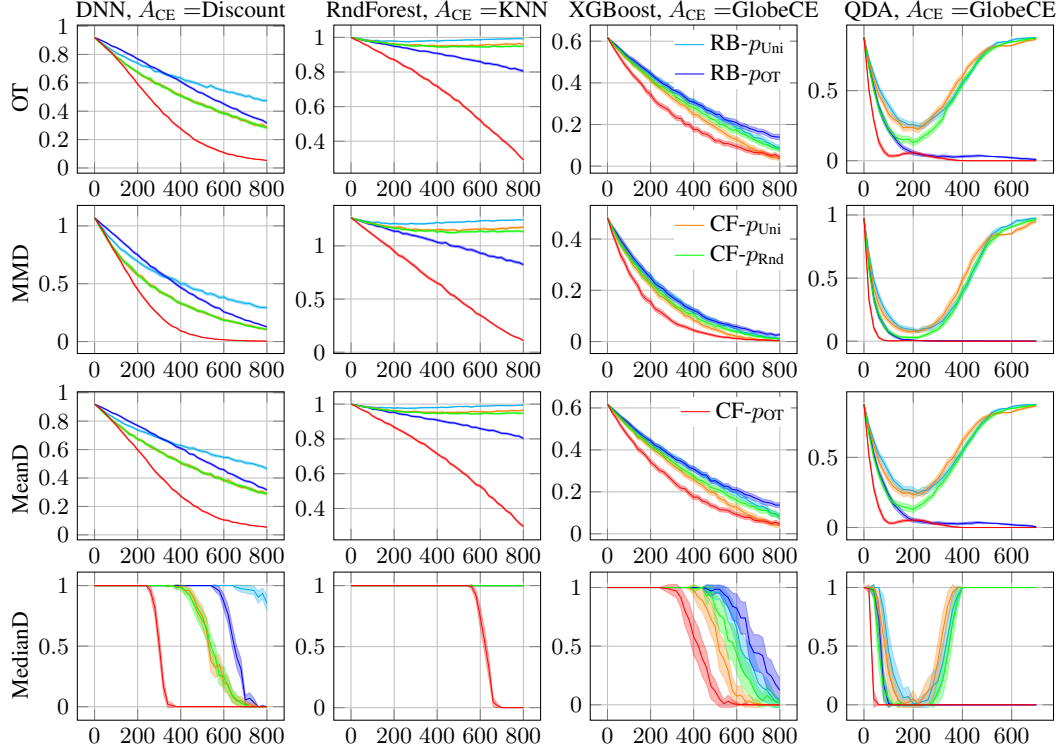


Figure 5: [HELOC] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of allowed actions C . Each experiment uses 100 runs, and shaded regions report 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

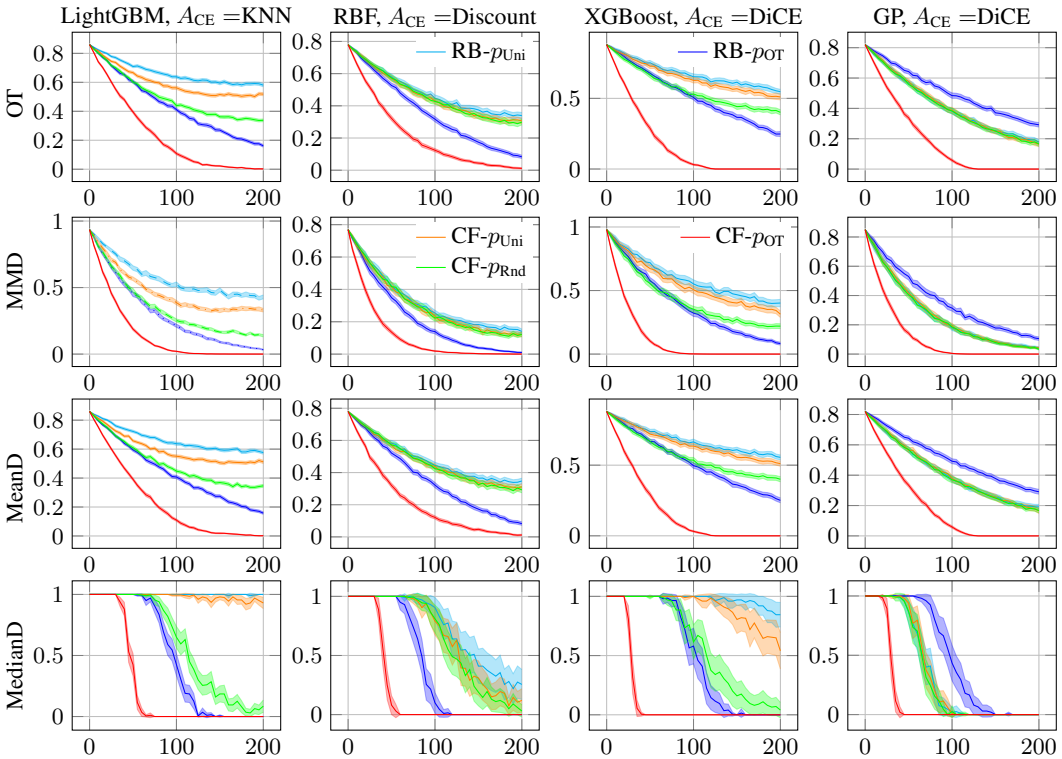


Figure 6: [German Credit] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of allowed actions C . Each experiment uses 100 runs, and shaded regions report 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

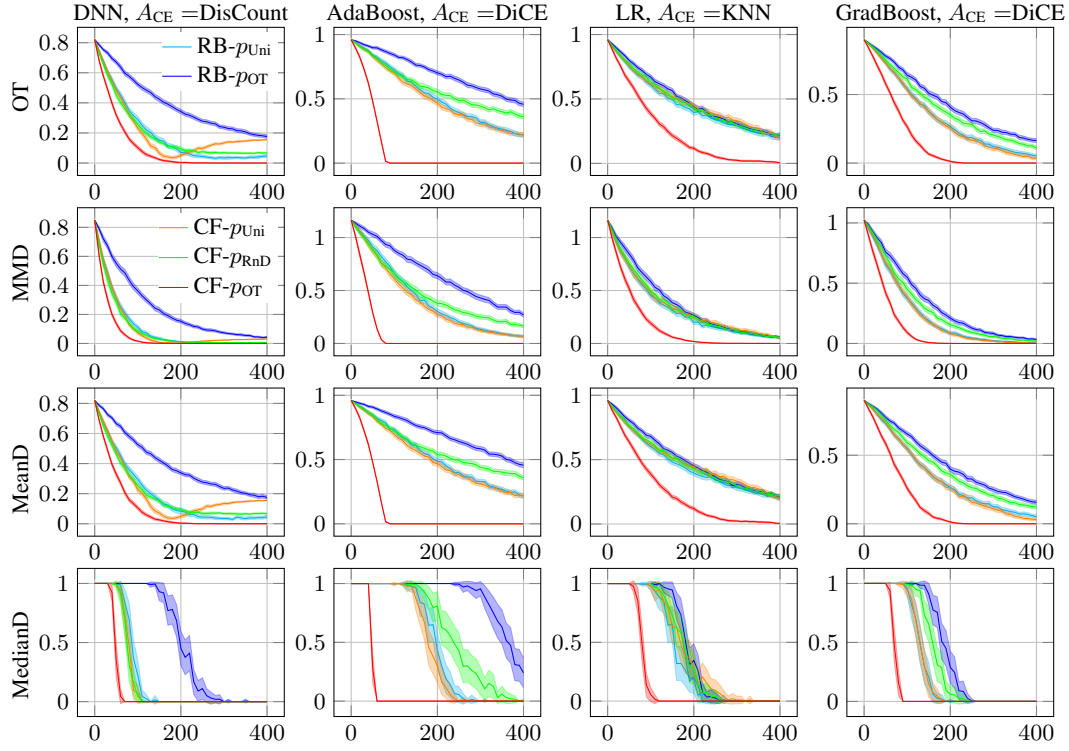


Figure 7: [Hotel Bookings] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of allowed actions C . Each experiment uses 100 runs, and shaded regions report 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.

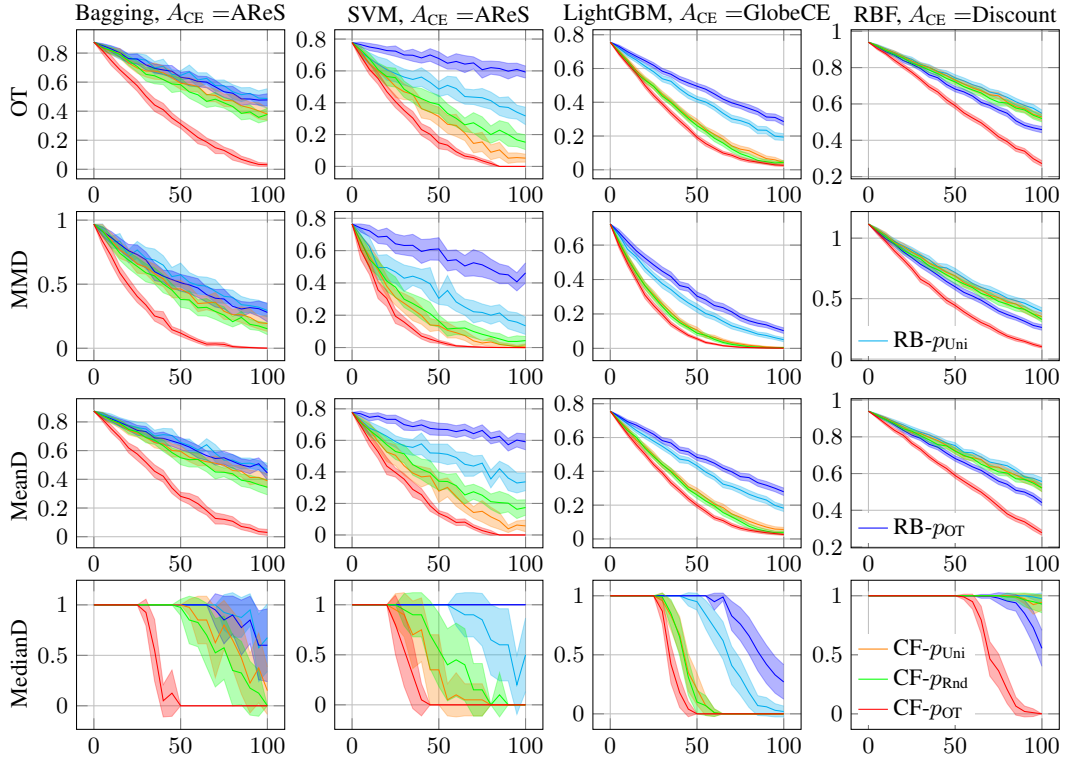


Figure 8: [COMPAS] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of allowed actions C . Each experiment uses 100 runs, and shaded regions report 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.